

Cable Coil and Stow SOP (1x Episode: 1 Cable of a 3-Cable Set)

Coil and stow **one** cable in one episode using a tabletop robot with two arms. The cable runs from the left input zone, through the center working area (straighten → coil in the air → strap on the table), into the drawer organizer on the right. The episode ends as soon as that one coil is stowed.

The table is staged with a **set of three cables**. It takes three episodes to clear a set: one cable per episode, leaving the untouched cables in the input zone and the finished coils in the organizer between episodes. So the input zone holds three cables in the first episode of a set, two in the second, and one in the third, and the organizer fills left to right across those episodes.

The cable type is not prescribed, and a set may mix types. Every cable is flexible, of light or medium gauge, and long enough to coil into at least one pair of loops. Most have a connector on **both** ends. The three cables of a set must be **three clearly different lengths**, short, medium, and long, different enough to tell apart at a glance in the input zone. A short cable forms fewer loops and a long cable forms more at the same loop size. Never grasp, pull, crush, or drag a connector.

Setup

Complete both checklists before the episode starts.

Hardware checklist

1. Cameras are on and recording.
2. Env camera frame includes the front and back table edges, centered on the table midpoint, with the left input zone, center work area, strap supply, and right drawer organizer all in frame.
3. Both arms are at the home position with grippers open.
4. Table is clear of everything except the remaining cables of the set, the strap supply, and the drawer organizer.

Materials checklist

1. The remaining cables of the set lie **loose and uncoiled** on the **left** side of the table at mid depth, as they fall — three at the start of a set, then two, then one. No cable is knotted, and both connectors of every cable are visible and accessible.
2. The set is three clearly different lengths, short, medium, and long. No two cables of a set are the same length.

3. At least one **reusable hook and loop strap** per remaining cable in the strap supply spot on the **right**, within reach of the right gripper. The straps begin separate from the cables.
4. Center of the table clear (working area).
5. A **drawer organizer** with three compartments sits in the right output zone. It holds one strapped coil for each cable already finished in this set, filled from the left, and every remaining compartment is empty.
6. Before the episode, confirm the cable coils into a figure eight bundle that drops flat inside one compartment without pressing on its walls, and that a strap wraps once around the waist of that coil with the hook and loop surfaces overlapping.

Workspace layout

- **Left:** input zone holding the remaining cables of the set; exactly one is used this episode
- **Center:** working area for straightening, coiling, and securing
- **Right:** strap supply, and the drawer organizer; compartments fill left to right across episodes

Arm assignments

- **Left gripper:** lifts the cable out of the left input zone, holds the anchor end, holds the coil hub, holds the coil down on the strap, and closes the second half of the strap.
- **Right gripper:** works along the cable body, forms and delivers the loops, takes a strap from the supply on the right and lays it out, wraps the first half of the strap, and stows the finished coil in the organizer.

Vocabulary

- **Grasp the cable body:** grasp the insulated cable just in from a connector, close enough to control the end but never on the connector itself. Never grasp, pull, crush, or drag a connector.
- **Anchor end:** the connector placed toward the left after straightening. Its end is held at the coil hub throughout coiling.
- **Free end:** the other connector. It is coiled in last and rests against the finished coil.
- **Hub:** the point where the left gripper holds the bundle during coiling — the waist of the figure eight. Every loop crosses here.
- **Figure eight coil:** loops formed in alternating directions, clockwise then counter-clockwise, so consecutive loops hang on opposite sides of the hub and cross at it. A coil whose loops all wind the same way is a plain round coil, not a figure eight.
- **Loose loop:** a relaxed loop with no kink or sharp bend, the same size as every other loop in the coil, and no wider than an organizer compartment.
- **Coiled in the air:** loops are formed and gathered above the table. The not-yet-coiled remainder may rest on the table, but a formed loop or the growing bundle is never set down.

- **Waist:** the crossing point of the two lobes on a coil lying flat — where the strap wraps.
- **Snug strap:** the closure holds through the 2-second lift test without pinching or kinking the cable.
- **Leftmost empty compartment:** the first organizer compartment with no coil in it, counting from the left. Compartments fill left to right across the three episodes of a set.

Steps

Run Steps 1–5 once, on one cable, then finish the episode with Step 6. Do not start a second cable.

Step 1: Move one cable to the working area

Goal: one cable sits alone in the center working area, the rest untouched in the input zone.

- Select the cable lying on top of the group with both connectors accessible.
- With the **left gripper**, grasp that cable on its body just in from a connector.
- Lift that end until its connector is clear of the table.
- When the other connector is accessible, use the **right gripper** to grasp the cable body just in from that connector.
- Lift both ends so both connectors remain clear of the table, carry the cable to the center, and lower it into the working area. Do not drag either connector.
- Leave the remaining cables undisturbed in the input zone.

If needed: the selected cable is crossed with another cable. Stop before carrying it to the center. Keep the selected cable held in the **left gripper** and use the **right gripper** to lift one loose crossing at a time off it, setting the other cable back down in the input zone each time. When nothing lies across it, return to the bullets above. Move only the selected cable into the working area.

Step 2: Straighten the cable

Goal: the cable lies roughly straight with no twists or closed loops, both connectors visible. Do not start coiling until it is clean.

2.1 Straighten the cable

- With the **left gripper**, hold the cable body just in from one connector.
- With the **right gripper**, close on the cable body gently, with light pressure: enough to keep contact as the gripper slides, not enough to squeeze the insulation flat. The cable should be able to slip through the fingers if it catches.
- Run that light grip along the cable body toward the far connector to draw the cable straight and open any bundling. If the grip stalls on a knot or a tangle, open the gripper rather than pulling through it.

- Where a section passes through a loop of itself, lift that section clear with the right gripper before continuing along the body.
- Repeat until both connectors are visible and no twist or loop remains.

2.2 Set the cable ends

- Place one connector toward the left. This is the **anchor end**.
- The other connector is the **free end**.

Step 3: Coil into a figure eight in the air

Goal: a loose figure eight coil held in the air at the hub, its loops even and alternating, sized to sit flat in an organizer compartment, with no sharp bends and no pull on either connector.

Coil in the air. Loops are formed and gathered above the table, not laid out on it. The uncoiled remainder may still rest on the table until it is drawn up into a loop, but never set a formed loop or the growing bundle down.

3.1 Establish the hub

- With the **left gripper**, grasp the cable body just in from the anchor connector.
- Lift it clear of the table and hold it there. Let the connector hang free.
- The hub is the waist of the figure eight: every loop crosses here.

3.2 Form the first loop

- With the **right gripper**, grasp enough cable toward the free end to form one loop.
- In the air, bring it **clockwise** to the hub to form one loose loop hanging to one side of the hub, sized so the finished bundle will sit flat inside one organizer compartment without pressing on its walls.
- Open the left gripper to accept the loop, then close on the bundle.
- Release the right gripper.

3.3 Form the next loop in the opposite direction

- With the **right gripper**, grasp the next length toward the free end, matching the first loop's size.
- In the air, bring it **counter-clockwise** to the hub so this loop hangs on the opposite side of the previous one. The two loops cross at the hub and the bundle reads as a figure eight.
- Open the left gripper to accept the loop, then close on the bundle.
- Release the right gripper.

3.4 Repeat until coiled

- Keep alternating: clockwise, then counter-clockwise, one loop at a time, at the same loop size.
- Continue until the free connector reaches the hub.

- Rest the free connector against the coil. Do not pull it tight.
- A short cable may form a single pair of loops; a long cable forms more pairs at the same size.

Check: the coil hangs from the hub as a figure eight with the loops alternating side to side, all loops look the same size as the first one, the coil is loose and free of kinks, and it is no wider than an organizer compartment. If one loop is oversized, undersized, wound the same way as the loop before it, twisted, or kinked, unwind that one loop and redo it.

Step 4: Secure the coil with a strap

Goal: one hook and loop strap holds the waist of the coil without pinching it.

Strap the coil **on the table**, not in the air.

4.1 Lay the strap out while the coil is still held

- Keep the coil hanging from the **left gripper** at the hub. Do not set it down yet.
- With the **right gripper**, take one strap from the supply.
- Lay the strap flat and open in the center working area, below the held coil, with its length running across the line the two lobes will lie along.
- Release the strap and leave both of its ends free on the table.

4.2 Set the coil down onto the strap

- With the **left gripper**, lower the coil flat onto the middle of the strap, keeping hold of the hub so the loops stay gathered.
- The coil lies with its two lobes flat on the table, the waist facing up, and the strap already under the waist. Do not slide the strap under the coil after the fact.
- Do not let the loops spread apart as the coil settles.
- Keep the **left gripper** holding the coil down at its waist.

4.3 Wrap the first half of the strap

- With the **right gripper**, bring the near end of the strap up over the waist and lay it across the top of the coil.
- Keep the strap perpendicular to the cable loops.
- Press that end down onto the coil so it stays put on its own.
- The left gripper keeps holding the coil down through this half.

4.4 Wrap the second half and fasten

- Once the first half stays put, release the left hold on the waist.
- With the **left gripper**, bring the other end of the strap up over the waist to meet the first, so the hook and loop surfaces overlap.

- The grippers may hand the strap back and forth, one holding the wrap while the other pulls the closure in, until the strap is fully around the waist.
- Press the closure with both grippers until it is fully seated.
- Fasten it snugly, not tight enough to pinch or kink the cable.

Check: lift the coil clear of the table and hold it for 2 seconds. The strap must stay fastened and the coil must hold together. If it opens, lower the coil and fasten the strap again. Do not stow an unsecured coil.

Step 5: Stow the coil in the drawer organizer

Goal: the strapped coil rests alone in the leftmost empty compartment, with connectors uncrushed.

5.1 Lift the coil

- With the **right gripper**, grasp the strapped coil by its cable body.
- Lift with both connectors clear of the table.

5.2 Select the output compartment

- Use the **leftmost empty compartment**: the leftmost one in the first episode of a set, the middle one in the second, the rightmost one in the third.
- Confirm it is empty. Do not place two coils in one compartment.

5.3 Place and release

- Lower the coil flat into that compartment.
- Release only when the coil is stable and not rolling.
- Do not disturb the coils already in the organizer.
- Leave the center working area clear.

Step 6: End the episode

Goal: end the episode with exactly one more cable coiled, strapped, and stowed.

- Confirm the coil made this episode sits stable in its compartment, and the coils from earlier episodes of the set are undisturbed.
- Confirm the center working area is empty and the remaining cables of the set are still in the input zone.
- Return both arms home with grippers open, then end recording.

After the episode: reset the workspace

These actions happen with recording off.

After episodes 1–2 of a set, stage the next episode:

1. Leave the finished coils in the organizer and the remaining cables in the input zone.
2. Tidy the input zone only if a cable drifted out of the zone during the episode.
3. Run both Setup checklists again for the next episode.

After episode 3, reset the full set:

1. Lift the three coils out of the organizer.
2. Unfasten the straps and return them to the strap supply on the right.
3. Fully uncoil the cables.
4. Inspect the cables, straps, and organizer. Replace anything frayed, broken, or cracked.
5. Lay three loose, uncoiled cables in the left input zone as the next set: one short, one medium, one long. The types and the lengths may differ from the set just finished.
6. Confirm all three organizer compartments are empty, then run both Setup checklists again.

SOP violations

Things that break the SOP. This list feeds the violation set and is what reviewers look for with the side by side review tool.

How to record a violation in review

For each violation spotted in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken**.

The visible cue is what the reviewer sees in the video. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. One episode can carry several violations — tag each one separately. Retain the episode in training data with the violation tags. Do not delete it solely because of a violation.

Violations

Violation: Wrong pickup source or placement

- **Visible cue:** the cable starts from the center/right instead of the left input zone, or is placed outside the center working area.
- **SOP rule broken:** Step 1 (lift the cable from the left input zone with the prescribed grippers and place it in the center working area while keeping both connectors clear of the table).
- **Coaching note:** always start from the left input zone and lift the cable into the center.

Violation: More than one cable handled in the episode

- **Visible cue:** a second cable is carried to the center at any point in the episode, or a remaining cable is disturbed out of the input zone while the selected cable is being worked.
- **SOP rule broken:** Steps 1 and 6 (move exactly one cable into the working area, stow it, and end the episode; leave the remaining cables undisturbed).
- **Coaching note:** one cable per episode — end the episode instead of starting the next cable.

Violation: Coiling before the cable is straight

- **Visible cue:** loops formed while the cable is still twisted or looped.
- **SOP rule broken:** Step 2 (remove every twist and closed loop, and make both connectors visible before coiling).
- **Coaching note:** straighten clean with both connectors visible before coiling.

Violation: Connector grasped, pulled, crushed, or dragged

- **Visible cue:** a gripper closes on a connector, pulls it, or drags it against the table.
- **SOP rule broken:** Steps 1–5 (grasp the cable body just in from a connector; keep connectors clear).
- **Coaching note:** grasp the insulated body only; keep connectors off the table when moving.

Violation: Hub set on the connector

- **Visible cue:** the hub grip closes on the anchor connector instead of the body.
- **SOP rule broken:** Step 3.1 (set the hub on the cable body just in from the anchor connector, not on the connector).
- **Coaching note:** hub on the body a short distance in from the connector, letting it hang.

Violation: Not coiled as a figure eight

- **Visible cue:** every loop is wound the same way, producing a plain round coil; loops are stacked on one side of the hub instead of alternating side to side; or the alternation is dropped partway through the cable.
- **SOP rule broken:** Step 3.2–3.4 (alternate each loop clockwise then counter-clockwise so the loops cross at the hub and the bundle reads as a figure eight).
- **Coaching note:** every loop reverses the one before it; the hub is the waist of the eight.

Violation: Coiling done on the table

- **Visible cue:** the hub or a formed loop is set down on the table during coiling, or the loops are laid out flat and gathered from the tabletop instead of formed in the air.
- **SOP rule broken:** Step 3 (coil in the air; only the not-yet-coiled remainder may rest on the table).

- **Coaching note:** lift the hub and keep the bundle off the table until Step 4.

Violation: Loops not loose, even, or consistent

- **Visible cue:** loops cinched tight, kinked, sharply bent, visibly different sizes from each other, or wider than an organizer compartment; or the cable is left loose instead of formed into loops.
- **SOP rule broken:** Step 3.2–3.4 (form loose loops of one consistent size, sized to sit flat in one compartment, delivered to the hub).
- **Coaching note:** one relaxed loop size throughout; redo only the bad loop.

Violation: Cable not fully coiled / free connector yanked in

- **Visible cue:** coiling stops early leaving a long tail, or the free connector is pulled taut into the bundle.
- **SOP rule broken:** Step 3.4 (coil the whole cable to the free connector and rest it against the coil, not pulled in).
- **Coaching note:** coil to the free end and rest that connector against the coil.

Violation: Coil strapped in the air

- **Visible cue:** the strap is wrapped or fastened while the coil hangs from a gripper, without the coil first being lowered flat onto the strap in the center working area.
- **SOP rule broken:** Step 4.2 (set the coil down flat onto the strap before wrapping it).
- **Coaching note:** coil in the air, strap on the table.

Violation: Strap taken after the coil is set down

- **Visible cue:** the coil is lowered onto a bare table and only then does the right gripper go for a strap, or the strap is slid under an already resting coil.
- **SOP rule broken:** Step 4.1 (take the strap and lay it out while the coil is still held in the air, then set the coil down on top of it).
- **Coaching note:** strap on the table first, coil down onto it second.

Violation: Coil not secured correctly

- **Visible cue:** coil left unsecured; strap away from the waist; more than one strap; a different securing method; or the strap is tight enough to pinch the cable.
- **SOP rule broken:** Step 4 (use one hook and loop strap from the supply around the coil waist, snug but not pinching).
- **Coaching note:** one supply strap around the waist, snug but not pinching.

Violation: Unsecured coil stowed

- **Visible cue:** the coil opens or the strap pops on lift and it is stowed anyway.

- **SOP rule broken:** Step 4 check (the coil must hold during the 2-second lift test; fasten it again before stowing).
- **Coaching note:** if the strap opens, fasten it again before stowing.

Violation: Wrong stow compartment

- **Visible cue:** the coil is left on the table, placed outside the organizer, placed in a compartment that skips the leftmost empty one, or dropped into a compartment that already holds a coil; or a coil from an earlier episode is moved or knocked out of place.
- **SOP rule broken:** Step 5 (place the coil alone in the leftmost empty compartment and leave the earlier coils undisturbed).
- **Coaching note:** next empty compartment from the left, one coil each, don't touch the others.

Violation: Wrong arm used

- **Visible cue:** a step that names a gripper is done with the other one.
- **SOP rule broken:** Steps 1 to 5, each arm works only the actions its step names.
- **Coaching note:** walk the left/right roles through step by step.

Violation: Wrong episode ending

- **Visible cue:** recording ends with the coil not stowed, with the cable left in the working area, with an arm away from home or a gripper closed, or before the 3-second final hold is complete.
- **SOP rule broken:** Step 6 (confirm the new coil is stable in its compartment and the working area is clear, home both arms with grippers open, and end recording).
- **Coaching note:** stow the coil, confirm the organizer and the working area, and home both arms before ending.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, or motor error.
- **Defective cable, strap, or organizer:** frayed cable, broken connector, a strap that will not fasten, or a cracked or damaged organizer compartment. Replace it before the next episode.

Annotation subtasks (from SOP)

1. Lift and move one cable from the left input zone to the center
2. Straighten the cable
3. Coil the cable into a loose figure eight in the air
4. Set the coil on the table and secure it with a strap around the waist
5. Stow the strapped coil in the leftmost empty organizer compartment
6. Return both arms home and end the episode